Senior Software Engineer – Solution Design Exercise

Background

The Department of Education is developing an **AI-powered knowledge base** to help students, teachers and corporate staff access and understand information such as processes, policies, guidelines, notices and other departmental documentation.

As part of this platform, documents from a variety of sources will need to be prepared before they can be effectively used by AI systems.

The Challenge

We would like you to provide your proposed **solution design for a Document Parsing capability**.

The goal of this capability is to take documents entering the platform and transform them into a form that is **AI-ready**.

We are deliberately keeping the problem open-ended. We would like you to determine what you believe the important problems, requirements and considerations are, and design a solution accordingly.

The goal of the exercise is **not to write code**. We are interested in understanding how you approach the problem, structure a solution, make engineering decisions and communicate your thinking.

There is no single correct solution.

Use of AI

You are welcome to use AI tools while completing this exercise.

If you use AI, please briefly declare:

- Which tools you used
- How you used them
- How you reviewed or validated their output

Use of AI will not count against you. This is also an opportunity to demonstrate how you use AI effectively as part of your engineering workflow.

Submission

Please submit your solution in a format that is easy to review, such as:

- A Markdown file
- A shared Google Doc

- A shared Notion document

Diagrams or other supporting material may be included if you feel they help communicate your solution.